

D440

Health and Wellness Through Nutritional Science

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A granola bar sold at a hospital café for \$2.50, with 200 mg of sodium, contains 20 g carbohydrate, 5 g protein, and 10 g fat. About how many calories does it provide?

- A. 35 kcal
- B. 140 kcal
- C. 100 kcal
- D. 190 kcal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A food provides 180 kcal per serving and contains 9 g of fat. What percentage of its calories come from fat?

- A. 9%
- B. 45%
- C. 20%
- D. 5%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

At a community wellness fair held every spring, a 34-year-old bus driver weighs 80 kg and is 1.60 m tall. He says he eats breakfast every day. What is his BMI and category?

- A. 31.3, overweight
- B. 50.0, obese
- C. 31.3, obese
- D. 128, obese

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A patient is 1.75 m tall and has a BMI of 24. About how much does the patient weigh?

- A. 7.8 kg
- B. 42.0 kg
- C. 24.0 kg
- D. 73.5 kg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A 50-year-old accountant who runs marathons, and who buys supplements online, has been taking a high-dose vitamin A supplement for a year. Why is the nurse concerned?

- A. Vitamin A is fat-soluble, so excess is stored and can build to toxic levels.
- B. Vitamin supplements are always safe at any dose.
- C. Vitamin A is water-soluble, so it is flushed out before it can help.
- D. Online supplements are the only ones that can cause toxicity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which vitamin, taken in excess, is mostly excreted in the urine rather than stored?

- A. Vitamin K
- B. Vitamin D
- C. Vitamin E
- D. Vitamin C

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A 29-year-old graphic designer, married last June, tells the nurse she and her partner plan to try for a baby in the next few months. What nutrition advice best fits current guidelines?

- A. No change is needed until the second trimester.
- B. Start folic acid once a pregnancy test is positive.
- C. Start about 400 mcg of folic acid daily now, before conception.
- D. Start an iron supplement now to prevent neural tube defects.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

The parent of an 8-month-old, who attends daycare three days a week, says, “She loves yogurt, so I want to switch her bottle to whole cow’s milk.” The parent is a dairy farmer. What should the nurse advise?

- A. Switch now and add an iron supplement.
- B. Keep giving breast milk or iron-fortified formula until about 12 months.
- C. Switching now is fine, because she already eats yogurt.
- D. Switch to skim milk instead of whole milk.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A patient with stage 4 chronic kidney disease who is not on dialysis, and who works nights as a security guard, asks for a snack. Which choice is best?

- A. Apple slices
- B. Salted potato chips
- C. A banana
- D. A glass of milk, because it is always healthy

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A patient with chronic kidney disease is told to avoid bananas, oranges, and potatoes. Which lab value is the main reason?

- A. Glucose
- B. Potassium
- C. Sodium
- D. Phosphorus

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A 22-year-old college athlete who follows a vegan diet, trains six days a week, and cooks for his roommates asks which plant food is a complete protein. Which is correct?

- A. None; plant foods can never supply complete protein.
- B. Plain white rice
- C. Tofu (soy)
- D. Peanut butter

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A newly diagnosed celiac patient, who is a high-school chemistry teacher, is choosing lunch in the hospital cafeteria. Her favorite food is pizza. Which meal is safe?

- A. Grilled chicken with rice and steamed vegetables
- B. Whole-wheat pasta, because whole grains are healthier
- C. Beef and barley soup
- D. A small slice of pizza, since a little gluten is fine

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A granola bar ~~sold at a hospital café for \$2.50, with 200 mg of sodium,~~ contains 20 g carbohydrate, 5 g protein, and 10 g fat. About how many calories does it provide?

Target word: total kcal **Decides:** Fat is 9 kcal/g; carbohydrate and protein are 4 kcal/g.

A	Wrong stem number	Adds the grams, not the calories.
B	Confusion set	Uses 4 kcal/g for fat too: $35 \text{ g} \times 4$.
C	Dropped element	Counts carbohydrate and protein but drops the fat.
D	Correct	$(20 \times 4) + (5 \times 4) + (10 \times 9) = 80 + 20 + 90 = 190$.

2. Answer B Concept 1 • transfer

A food provides 180 kcal per serving and contains 9 g of fat. What percentage of its calories come from fat?

Target word: solve for percent from fat **Decides:** $9 \text{ g} \times 9 \text{ kcal/g} = 81 \text{ kcal}$; $81 \div 180$.

A	Wrong stem number	The gram figure, not a percentage of calories.
B	Correct	$81 \div 180 = 0.45$.
C	Confusion set	Uses 4 kcal/g for fat: $36 \div 180$.
D	Dropped element	Divides grams by calories without converting grams to kcal.

3. Answer C Concept 2

At a community wellness fair ~~held every spring,~~ a 34-year-old bus driver weighs 80 kg and is 1.60 m tall. ~~He says he eats breakfast every day.~~ What is his BMI and category?

Target word: BMI value and class **Decides:** $80 \div (1.60 \times 1.60) = 80 \div 2.56$.

A	Adjacent category	Right number, wrong band; overweight ends at 29.9.
B	Dropped element	Divides by height without squaring it.
C	Correct	$80 \div 2.56 = 31.25$, which is 30 or above.
D	Inverted operation	Multiplies weight by height instead of dividing.

4. Answer D Concept 2 • transfer

A patient is 1.75 m tall and has a BMI of 24. About how much does the patient weigh?

Target word: solve for weight **Decides:** $\text{Weight} = \text{BMI} \times \text{height}^2 = 24 \times 3.0625$.

A	Inverted operation	Divides BMI by height squared instead of multiplying.
B	Dropped element	Multiplies by height without squaring it.
C	Wrong stem number	Uses the BMI as the weight.
D	Correct	$24 \times 1.75^2 = 24 \times 3.0625 = 73.5 \text{ kg}$.

5. Answer A Concept 3

A 50-year-old accountant ~~who runs marathons, and who buys supplements online,~~ has been taking a high-dose vitamin A supplement for a year. Why is the nurse concerned?

Target word: toxicity risk **Decides:** Vitamin A is fat-soluble and stored in the liver.

A	Correct	Fat-soluble vitamins accumulate rather than being excreted.
B	Absolute language	Fat-soluble vitamins can reach toxic levels.
C	Inverted operation	Reverses the category; vitamin A is fat-soluble.
D	Camouflage	Where it was bought does not decide the risk; the vitamin type does.

6. Answer D Concept 3 • transfer

Which vitamin, taken in excess, is mostly excreted in the urine rather than stored?

Target word: reverse: find the water-soluble one **Decides:** Water-soluble vitamins are excreted.

A	Adjacent category	Fat-soluble; also matters for patients on warfarin.
B	Confusion set	Fat-soluble and stored; excess can cause high calcium.
C	Inverted operation	Fat-soluble and stored, the opposite of excreted.
D	Correct	Vitamin C is water-soluble; excess leaves in urine.

7. Answer C Concept 4

A 29-year-old ~~graphic designer, married last June,~~ tells the nurse she and her partner plan to try for a baby in the next few months. What nutrition advice best fits current guidelines?

Target word: timing and nutrient **Decides:** Folate must be in place before the neural tube closes.

A	Dropped element	Drops the early-closure fact.
B	Inverted operation	Reverses the timing; waiting may miss the window.
C	Correct	The neural tube closes early, often before pregnancy is known.
D	Adjacent category	Iron prevents anemia, not neural tube defects.

8. Answer B Concept 5

The parent of an 8-month-old, ~~who attends daycare three days a week,~~ says, "She loves yogurt, so I want to switch her bottle to whole cow's milk." ~~The parent is a dairy farmer.~~ What should the nurse advise?

Target word: milk choice before 12 months **Decides:** Cow's milk is delayed until about 12 months.

A	Dropped element	Still drops the 12-month guideline.
B	Correct	Cow's milk is low in iron and hard on an infant's kidneys before one year.
C	Camouflage	Leans on the yogurt detail; small dairy foods are not the same as replacing formula.
D	Adjacent category	Changes the milk type but keeps the age problem; skim is also wrong for infants.

9. Answer A Concept 6

A patient with stage 4 chronic kidney disease who is not on dialysis, ~~and who works nights as a security guard,~~ asks for a snack. Which choice is best?

Target word: best CKD snack **Decides:** Low potassium, phosphorus, and sodium.

A	Correct	Apples are low in potassium, phosphorus, and sodium.
B	Dropped element	Drops the sodium and potassium limits.
C	Confusion set	Healthy for many people but high in potassium.
D	Absolute language	Milk is high in phosphorus and potassium.

10. Answer B Concept 6 • transfer

A patient with chronic kidney disease is told to avoid bananas, oranges, and potatoes. Which lab value is the main reason?

Target word: reverse: from food limit to lab **Decides:** These foods are high in potassium.

A	Inverted operation	These limits are not about blood sugar.
B	Correct	Failing kidneys cannot clear potassium, and high levels threaten the heart.
C	Adjacent category	Sodium comes mainly from salt and processed foods.
D	Confusion set	Also limited in CKD, but dairy and colas are the main sources.

11. Answer C Concept 7

A 22-year-old college athlete who follows a vegan diet, ~~trains six days a week, and cooks for his roommates,~~ asks which plant food is a complete protein. Which is correct?

Target word: complete plant protein **Decides:** Soy supplies all essential amino acids.

A	Absolute language	Soy, quinoa, and grain-legume pairings do.
B	Dropped element	Grains are low in lysine unless paired with legumes.
C	Correct	Soy is one of the few plant foods that is a complete protein.
D	Confusion set	A good protein source, but not complete on its own.

12. Answer A Concept 8

A newly diagnosed celiac patient, ~~who is a high school chemistry teacher,~~ is choosing lunch in the hospital cafeteria. ~~Her favorite food is pizza.~~ Which meal is safe?

Target word: gluten-free choice **Decides:** Avoid wheat, barley, and rye.

A	Correct	None of these contain wheat, barley, or rye.
B	Adjacent category	Whole wheat is still wheat.
C	Confusion set	Barley contains gluten.
D	Absolute language	Even small amounts damage the intestine in celiac disease.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Energy per gram	Carbohydrate and protein give 4 kcal/g, fat gives 9 kcal/g, and alcohol gives 7 kcal/g.	Using 4 kcal/g for every macronutrient, or counting grams as calories.	A label listing grams of carbohydrate, protein, and fat.
2	Body mass index	BMI = weight (kg) ÷ height (m) ² ; 18.5–24.9 normal, 25–29.9 overweight, 30 and above obese.	Dividing by height without squaring it, or mixing pounds and meters.	Weight in kilograms and height in meters in the same stem.
3	Fat- vs. water-soluble vitamins	Vitamins A, D, E, and K are fat-soluble and stored, so excess can build to toxic levels; B vitamins and C are water-soluble and excess is mostly excreted.	Believing all vitamin excess is harmless because it is “natural.”	High-dose supplements or a question about toxicity or storage.
4	Folate before and during pregnancy	Folic acid (about 400 mcg daily) should start before conception to lower neural tube defect risk, because the neural tube closes in the first weeks.	Waiting until pregnancy is confirmed or late in the first trimester.	A person planning pregnancy or in early pregnancy.
5	Infant feeding	Infants get breast milk or iron-fortified formula for the first year; cow’s milk is delayed until about 12 months.	Switching to cow’s milk early because it seems wholesome.	An infant under 12 months and a question about milk.
6	Kidney disease diet	Chronic kidney disease diets limit sodium, potassium, and phosphorus because failing kidneys cannot clear them.	Thinking “healthy” high-potassium foods like bananas and oranges are always good choices.	CKD plus a food-choice question.
7	Complete and complementary proteins	A complete protein supplies all essential amino acids; soy and quinoa are complete plant proteins, and pairing grains with legumes also covers them.	Believing plant diets cannot supply complete protein.	A vegetarian or vegan patient and protein adequacy.
8	Gluten-free diet for celiac disease	Celiac disease requires lifelong avoidance of wheat, barley, and rye; rice, corn, and potatoes are safe.	Thinking whole grain or “low wheat” products are acceptable.	A celiac diagnosis and a menu choice.

Course scope (public description): The health and wellness through nutritional science course prepares nursing students to learn the basic principles of nutrition, nutrition throughout the life cycle, nutrition related to weight management and physical health, and nutrition related to patient conditions. Students will learn how nutrition influences a patient's overall health status across the life span.